

Data Flow Diagram

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Team ID	SWTID-2026-8096
Project Name	Rising Waters
Maximum Marks	2 Marks

Data Flow Diagram

- The Data Flow Diagram (DFD) for the Rising Waters – AI-Based Flood Prediction System illustrates how rainfall data flows through the system to generate flood predictions.
- The process begins when the user enters annual and seasonal rainfall values through the web interface.
- The system first validates the input data to ensure that all values are complete and within acceptable ranges.
- The validated data is then sent to the Flood Prediction process, where the trained Machine Learning model uses historical rainfall data stored in the Rainfall Dataset and Trained ML Model data store to analyze the inputs and predict the likelihood of flooding.
- Finally, the system displays the prediction result—either Flood Chance or No Flood Chance—along with relevant safety recommendations.

Steps:

1. User enters annual and seasonal rainfall values.
2. System checks and validates the entered data.
3. Machine learning model analyzes the data and predicts flood risk.
4. System displays **Flood Chance** or **No Flood Chance** with safety tips.

